

Farm Animal Demonstration

Title: Farm Animal Details using Classes and Objects

Objective /Aim:

To design and implement a Java program using classes and objects to store and display the details of different farm animals such as their name, place, food, and sound

System Requirements :

- **Hardware**
 - Standard PC/Laptop
 - Intel Core i3/i5 processor or equivalent
 - Minimum 4 GB RAM
 - Minimum 500 MB free storage
 - Keyboard and Monitor
- **Software**
 - Operating System: Windows / Linux / macOS
 - Java Development Kit (JDK 17 or JDK 21)
 - Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code
 - Command Prompt / Terminal

Algorithm / Logic Description :

1. Create a class named FarmAnimal.
2. Define attributes such as:
 - Animal name
 - Place where the animal stays
 - Food eaten by the animal
 - Sound made by the animal
3. Create a constructor to initialize the animal details.
4. Define a display() method to display the animal details.
5. Create the FarmAnimalDemo class containing the main() method.
6. Create a cow object with the details:

- Name: Cow
- Place: Cowshed
- Food: Grass
- Sound: Moo

7. Create a pig object with the details:

- Name: Pig
- Place: Pigsty
- Food: Vegetables
- Sound: Oink

8. Create a horse object with the details:

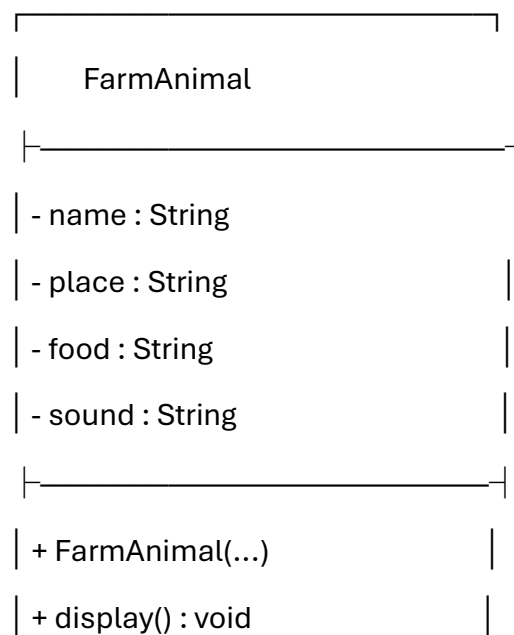
- Name: Horse
- Place: Stable
- Food: Grass
- Sound: Neigh

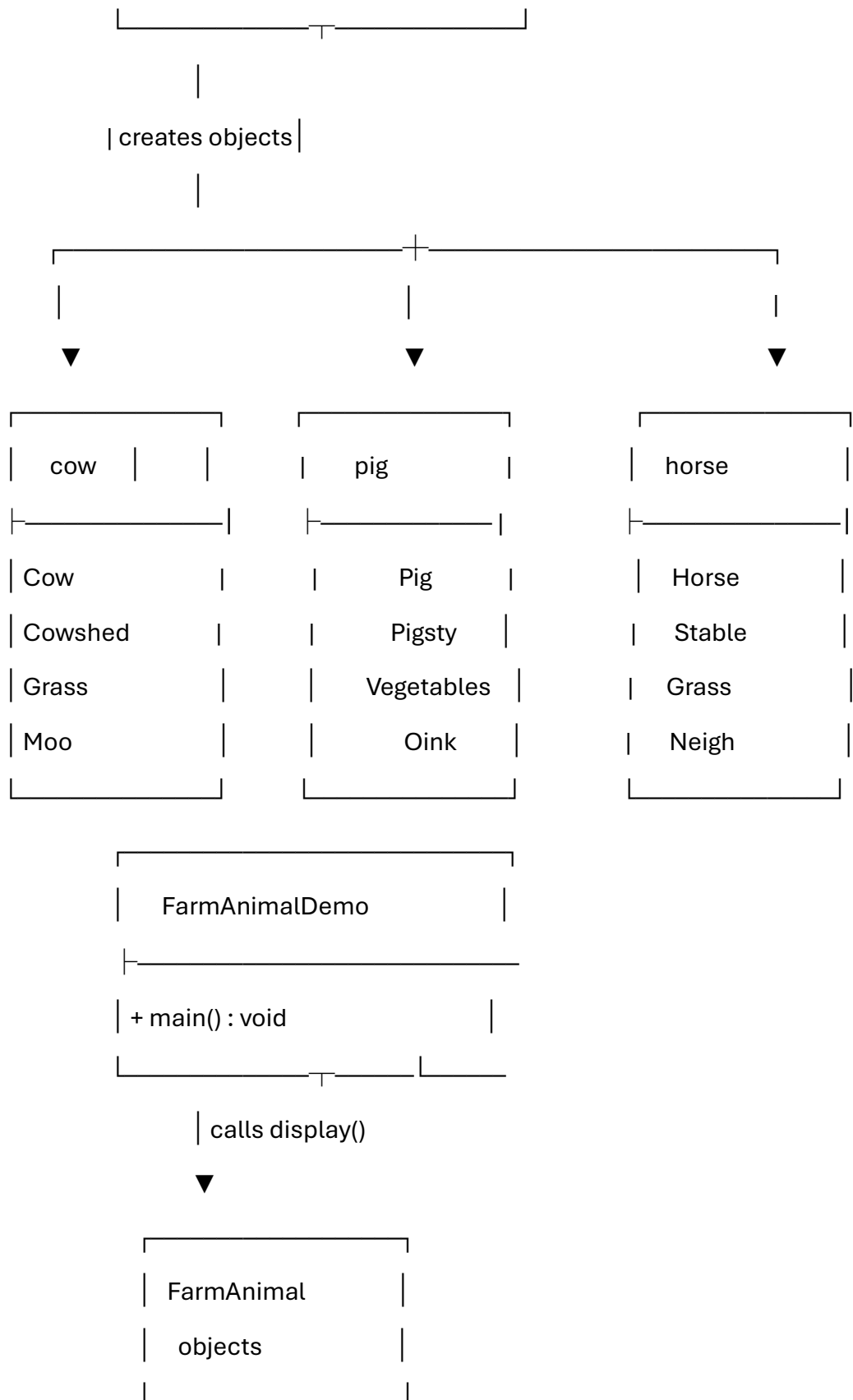
9. Call the display() method for each object.

10. Display the details of all three animals.

11. Stop the program.

UML class Diagram





Relationship: `FarmAnimalDemo` creates objects of the `FarmAnimal` class. The `cow`, `pig`, and `horse` objects are instances of the `FarmAnimal` class.

Source Code :

```
package core_java;

class FarmAnimal {

    String name;

    String place;

    String food;

    String sound;

    // Constructor to initialize animal details

    FarmAnimal(String name, String place, String food, String sound) {

        this.name = name;

        this.place = place;

        this.food = food;

        this.sound = sound;

    }

    // Method to display animal details

    void display() {

        System.out.println("Animal Name : " + name);

        System.out.println("Stays At  : " + place);

        System.out.println("Eats      : " + food);

        System.out.println("Sound   : " + sound);

        System.out.println();

    }

}

public class FarmAnimalDemo {

    public static void main(String[] args) {

        // Creating a cow object

        FarmAnimal cow = new FarmAnimal( "Cow", "Cowshed", "Grass", "Moo");

        // Creating a pig object
```

```
FarmAnimal pig = new FarmAnimal( "Pig", "Pigsty", "Vegetables", "Oink" );  
  
// Creating a horse object  
  
FarmAnimal horse = new FarmAnimal( "Horse", "Stable", "Grass", "Neigh" );  
  
// Displaying details of each animal  
  
cow.display();  
  
pig.display();  
  
horse.display();  
  
}  
  
}
```

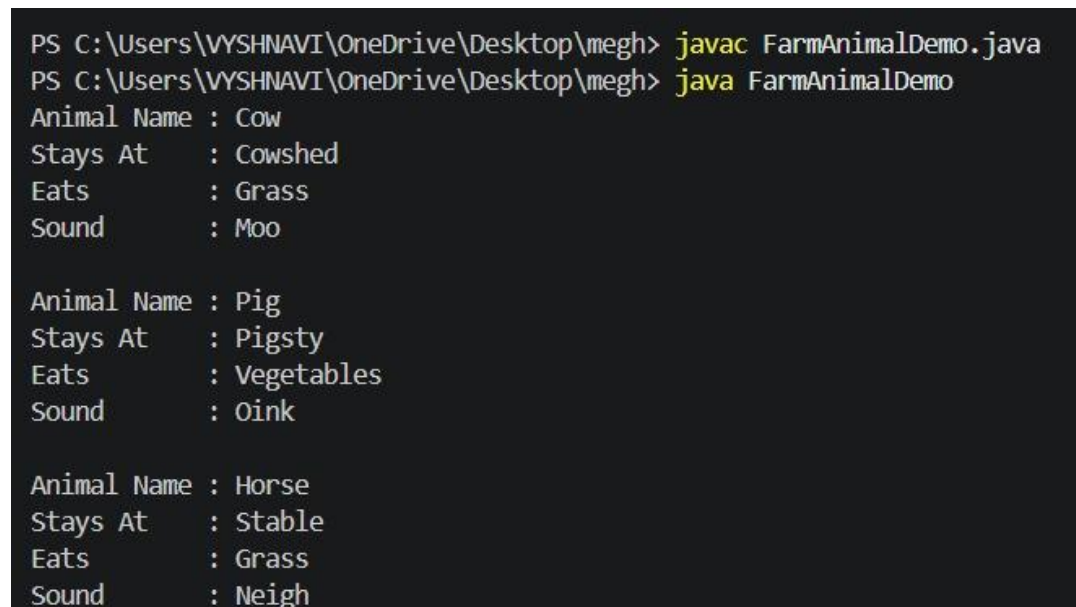
Compilation & Execution Commands:

Save file as : FarmAnimalDemo.java

Compilation: javac FarmAnimalDemo.java

Execution: java FarmAnimalDemo

Output Screenshot:



```
PS C:\Users\VYSHNAVI\OneDrive\Desktop\megh> javac FarmAnimalDemo.java  
PS C:\Users\VYSHNAVI\OneDrive\Desktop\megh> java FarmAnimalDemo  
Animal Name : Cow  
Stays At    : Cowshed  
Eats        : Grass  
Sound       : Moo  
  
Animal Name : Pig  
Stays At    : Pigsty  
Eats        : Vegetables  
Sound       : Oink  
  
Animal Name : Horse  
Stays At    : Stable  
Eats        : Grass  
Sound       : Neigh
```

Result :

The Farm Animal Details program was successfully implemented in Java using classes and objects. The program demonstrates the use of a constructor, object creation, data members, and methods to initialize and display the details of different farm animals.

